

Machine Learning Principles

Class 20: November 13

Gaussian Mixture Modeling and EM algorithm

Instructor: Diana Kim

Today's Lecture

1. Gaussian Mixture Modeling (GMM)

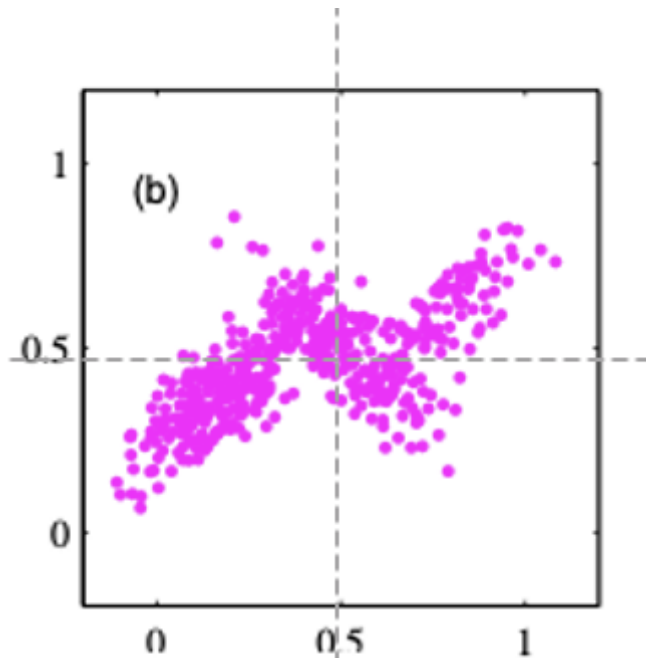
- DAG structure & CPT
- learning GMM using EM

2. Clustering

- K-means clustering (Comparison to EM of GMM)
- DBSCAN (Density Based Spatial Clustering of Application with Noise)
- HDBSCAN
: Hierarchical Density Based Spatial Clustering of Application with Noise

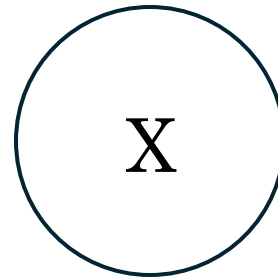
[**] GMM (hierarchical structure with a latent variable)

which model is structured so efficient and scalable?
Is the second model, Z accessible?



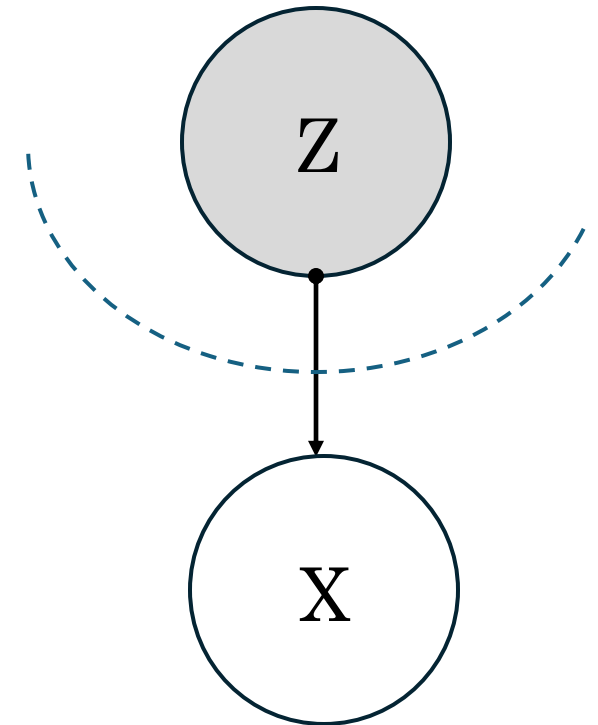
$P(x)$

from textbook Bishop Fig. 9.5



[without Z]

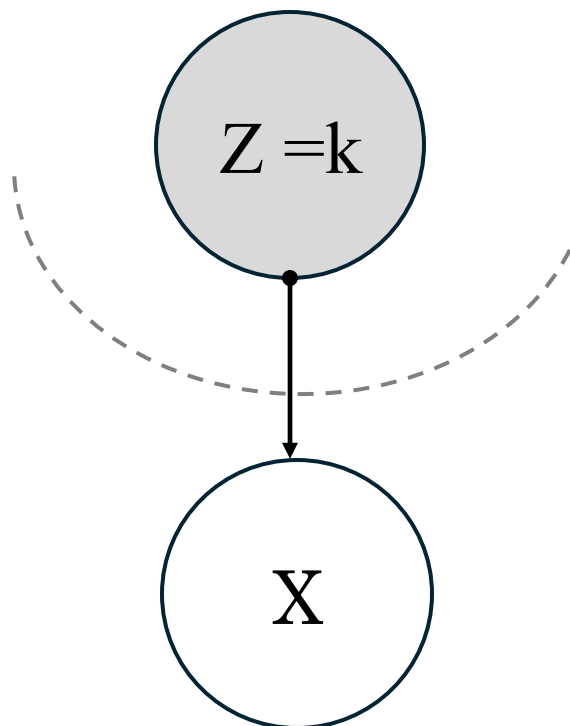
vs.



[with Z]

** GMM (parameters)

■ Bayesian Network Representation of GMM



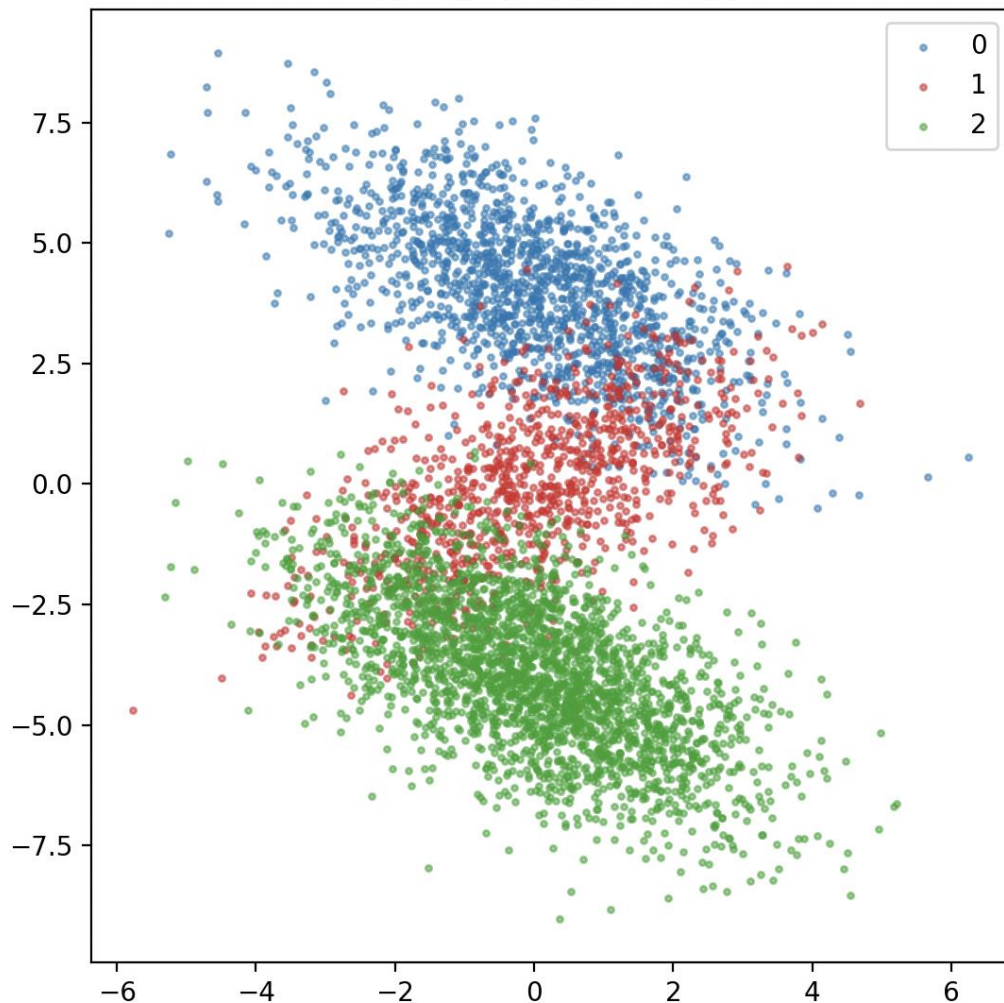
- $P(Z = k) = \pi_k$

- $P(X|Z = k) = \mathcal{N}(\mu_k, \Sigma_k)$

$$\begin{aligned} P(X) &= \sum_{Z=k} P(X, Z) = \sum_{Z=k} P(Z = k)P(X|Z = k) \\ &= \sum_{Z=k} \pi_k \mathcal{N}(\mu_k, \Sigma_k) \end{aligned}$$

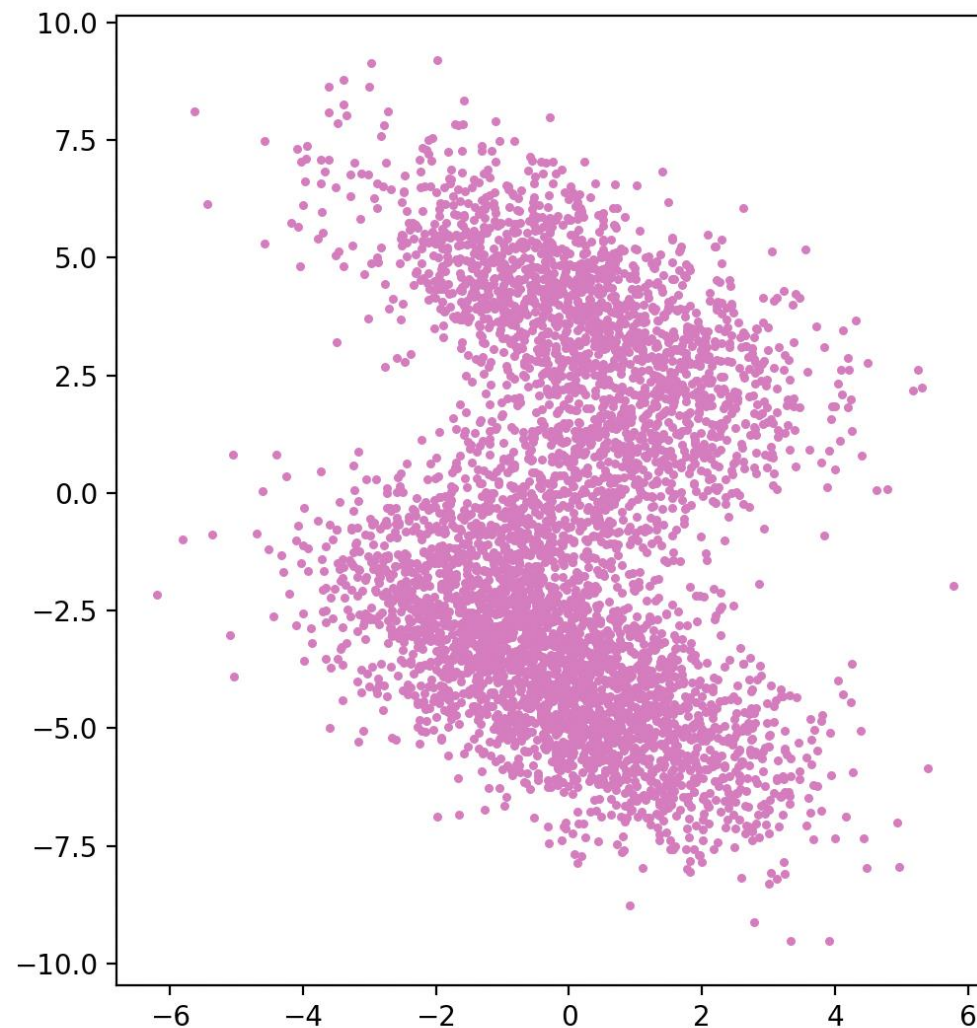
[**] GMM (generation example)

Gaussian Mixture Model



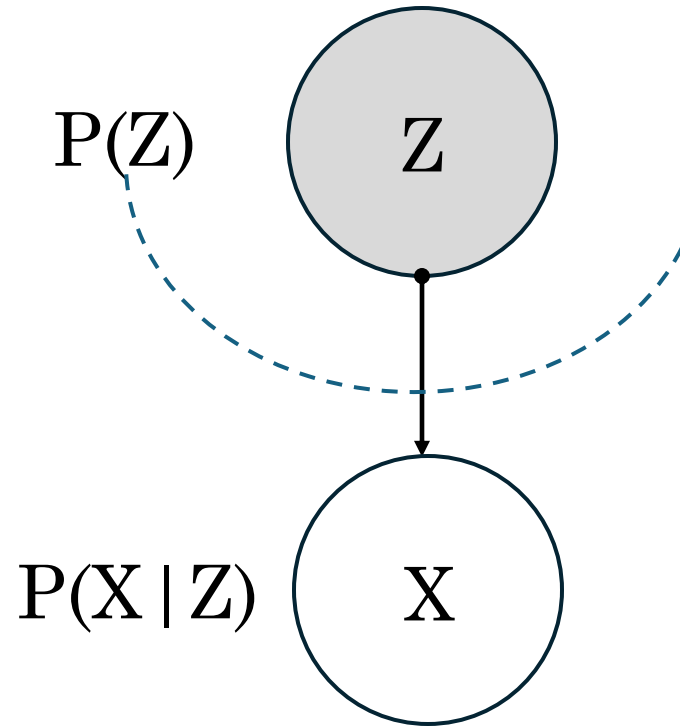
[when z is known]

Gaussian Mixture Model



[when z is unknown]

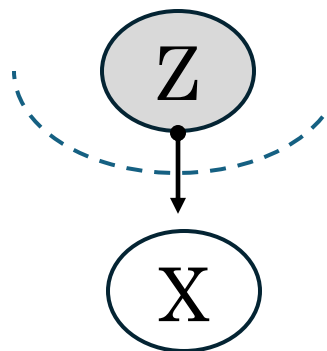
[**] GMM (data compression/representation)



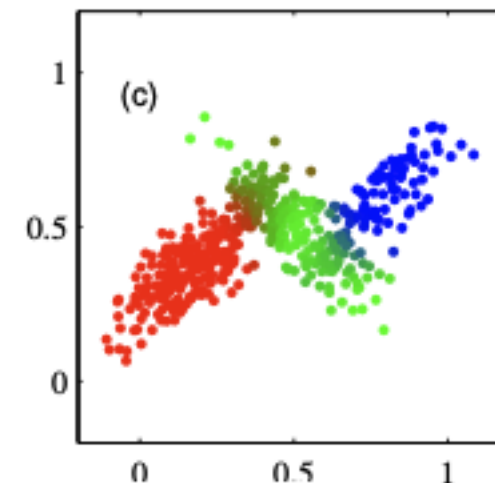
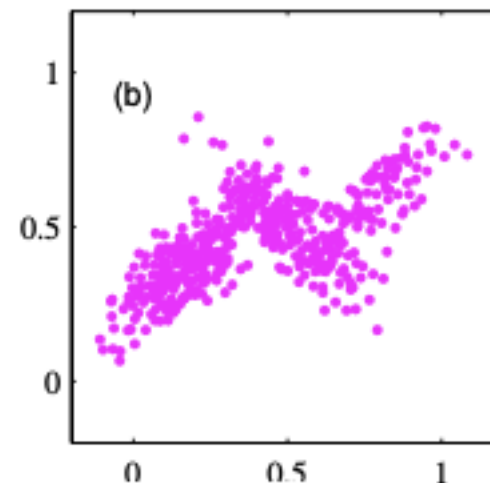
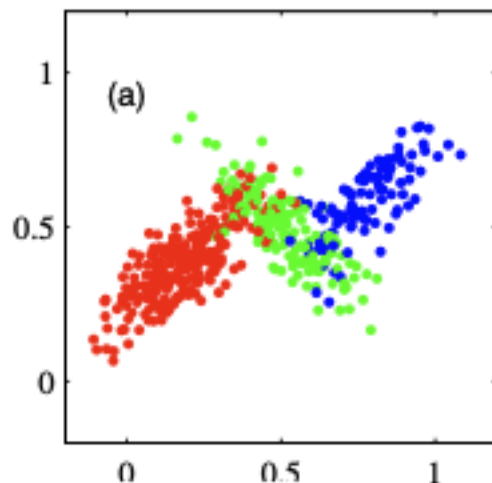
once we set this model,
we could compute the followings.

- $P(X)$
- $P(Z = k | x)$
 - * valuable information
 - * prediction about invisible factors.
 - * useful for clustering (when z is discrete)

[**] GMM (data compression/representation)



$$P(Z, X) = P(Z) \cdot P(X|Z)$$



$P(Z|x)$

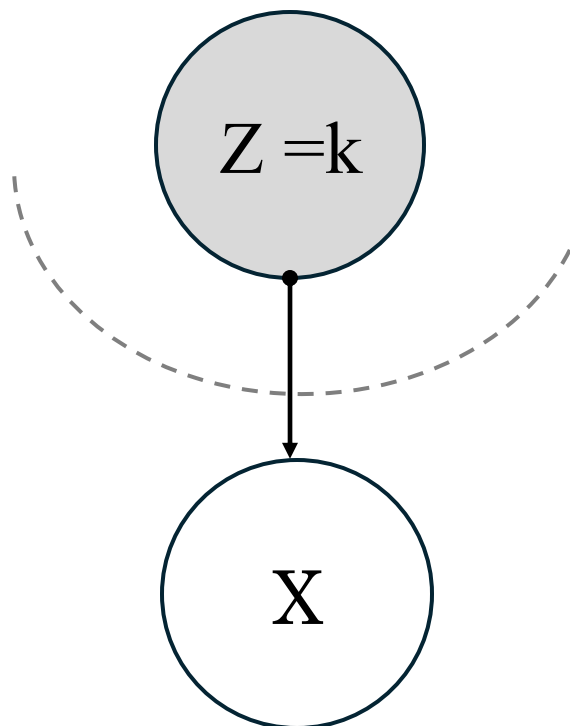
- Representation will be $Z = \operatorname{argmax} P(Z|x)$
- the representation will be one of colors: red / green / blue.

from textbook Bishop Fig. 9.5

- EM for GMM: “Given a data set”, how can we learn the parameters?

** GMM (parameters)

■ Bayesian Network Representation of GMM



- $P(Z = k) = \pi_k$

- $P(X|Z = k) = \mathcal{N}(\mu_k, \Sigma_k)$

$$\begin{aligned} P(X) &= \sum_{Z=k} P(X, Z) = \sum_{Z=k} P(Z = k)P(X|Z = k) \\ &= \sum_{Z=k} \pi_k \mathcal{N}(\mu_k, \Sigma_k) \end{aligned}$$

** GMM

Q: Learning a GMM:

is to learn the parameters : π, μ, Σ (as structure is given)
but we cannot access Z (imaginary / invisible).

how can we learn the parameters?
what method can we use?

[1] EM for GMM (likelihood & lower bound)

$$\begin{aligned}\log P(D|\theta) &= \sum_{n=1}^N \log \sum_{Z_n=k} P(Z_n, X_n|\theta) \\&= \sum_{n=1}^N \log \sum_{Z_n=k} \frac{P(Z_n, X_n|\theta)q(Z_n = k)}{q(Z_n = k)} \\&= \sum_{n=1}^N \log E\left[\frac{P(Z_n, X_n|\theta)}{q(Z_n = k)}\right] \quad \blacksquare \text{ expectation over } q(Z_n = k) \\&\geq \sum_{n=1}^N E\left[\log\left(\frac{P(Z_n, X_n|\theta)}{q(Z_n = k)}\right)\right] \\&= \sum_{n=1}^N \sum_k q(Z_n = k) \cdot (\log \pi_k + \log \mathcal{N}(x_n|\mu_k, \Sigma_k)) + \sum_{n=1}^N H(q(Z_n = k))\end{aligned}$$

Q: what $q(z_n = k)$ gives a tight bound?

[**] Learning Bayesian Tree (partial data observation)

Q: what density makes the lower bound tight?

$$\begin{aligned} & \sum_{n=1}^N E[\log P(S_n, R_n, C_n | \theta) + H(q(C_n))] \\ &= \sum_{n=1}^N \sum_{C_n} q(C_n) \log \left(\frac{P(S_n, R_n, C_n | \theta)}{q(C_n)} \right) \\ &= \sum_{n=1}^N \sum_{C_n} q(C_n) \log \left(\frac{P(C_n | S_n, R_n, \theta) P(S_n, R_n | \theta)}{q(C_n)} \right) \\ &= \sum_{n=1}^N \left(\sum_{C_n} q(C_n) \log \frac{P(C_n | S_n, R_n, \theta)}{q(C_n)} + \sum_{C_n} q(C_n) \log P(S_n, R_n | \theta) \right) \\ &= \sum_{n=1}^N -KL(q(C_n) || P(C_n | S_n, R_n, \theta)) + \log P(S_n, R_n | \theta) \end{aligned}$$

when $q(C_i) = P(C_i | S_i, R_i, \theta, p, q)$

the lower bound becomes equal to $\log P(S_i, R_i | \theta, p, q)$

[2] EM for GMM (Expectation E step)

parameters at time t

$$\begin{aligned} q(Z_n = k) &= P(Z_n = k | X_n = x_n, \pi_k(t), \mu_k(t), \Sigma_k(t)) \\ &= \frac{P(Z_n = k, X_n = x_n | \pi_k(t), \mu_k(t), \Sigma_k(t))}{P(X_n = x_n | \pi_t, \mu_t, \Sigma_t)} \\ &= \frac{P(Z_n = k | \pi_k(t), \mu_k(t), \Sigma_k(t)) P(X_n = x_n | Z_n = k, \pi_k(t), \mu_k(t), \Sigma_k(t))}{P(X_n = x_n | \pi_k(t), \mu_k(t), \Sigma_k(t))} \\ \gamma_{nk}(\pi_k(t), \mu_k(t), \Sigma_k(t)) &= \frac{P(Z_n = k | \pi_t, \mu_t, \Sigma_t) P(X_n = x_n | Z_n = k, \pi_k(t), \mu_k(t), \Sigma_k(t))}{P(X_n = x_n | \pi_k(t), \mu_k(t), \Sigma_k(t))} \end{aligned}$$

- **responsibility** that cluster K takes for data point n

- M step
- [1] update μ_k

$$\sum_{n=1}^N \sum_k q(Z_n = k) \cdot (\log \pi_k + \log \mathcal{N}(x_n | \mu_k, \Sigma_k)) + \sum_{n=1}^N H(q(Z_n = k))$$

+ this is the lower bound we computed in the previous slide.

$$L = \sum_{n=1}^N \sum_k q(Z_n = k) \cdot (\log \pi_k + \log \mathcal{N}(x_n | \mu_k, \Sigma_k)) + \sum_{n=1}^N H(q(Z_n = k))$$

$$L = \sum_{n=1}^N \sum_k r_{nk} \cdot (\log \pi_k + \log \mathcal{N}(x_n | \mu_k, \Sigma_k)) + \sum_{n=1}^N H(r_{nk})$$

$$\frac{\partial L}{\partial \mu_k} = \frac{\partial}{\partial \mu_k} \sum_{n=1}^N (-1/2 \gamma_{nk} (x_n - \mu_k)^t \Sigma^{-1} (x_n - \mu_k))$$

$$\frac{\partial L}{\partial \mu_k} = \sum_{n=1}^N \gamma_{nk} \Sigma^{-1} (x_n - \mu_k) = 0$$

$$\mu_k = \frac{\sum_{n=1}^N \gamma_{nk} x_n}{\sum_{n=1}^N \gamma_{nk}}$$

$\mathbf{y}$	$\frac{\partial \mathbf{y}}{\partial \mathbf{x}}$
$\mathbf{Ax}$	$\mathbf{A}^T$
$\mathbf{x}^T \mathbf{A}$	$\mathbf{A}$
$\mathbf{x}^T \mathbf{x}$	$2\mathbf{x}$
$\mathbf{x}^T \mathbf{Ax}$	$\mathbf{Ax} + \mathbf{A}^T \mathbf{x}$

- M step of GMM
- [2] update Σ_k

$$L = \sum_{n=1}^N \sum_k r_{nk} \cdot (\log \pi_k + \log \mathcal{N}(x_n | \mu_k, \Sigma_k)) + \sum_{n=1}^N H(r_{nk})$$

$$\frac{\partial L}{\partial \Sigma_k^{-1}} = \frac{\partial}{\partial \Sigma_k^{-1}} \sum_{n=1}^N (-1/2 (\gamma_{nk} (x_n - \mu_k)^t \Sigma^{-1} (x_n - \mu_k) - \log \sqrt{(2\pi)^n \cdot 1/|\Sigma^{-1}|}))$$

$$= \sum_{n=1}^N -1/2 \gamma_{nk} (x_n - \mu_k)^t (x_n - \mu_k) + \gamma_{nk} 1/2 \cdot \Sigma_k = 0$$

$$\Sigma_k = \frac{\sum_{n=1}^N \gamma_{nk} (x_n - \mu_k)^t (x_n - \mu_k)}{\sum_{n=1}^N \gamma_{nk}}$$

$$\frac{\partial}{\partial A} \log |A| = A^{-T}$$

$$\frac{\partial}{\partial A} x^T A x = \frac{\partial}{\partial A} \text{tr}[x x^T A] = [x x^T]^T = x x^T$$

- M step of GMM
[3] update π_k

$$L = \sum_{n=1}^N \sum_k r_{nk} \cdot (\log \pi_k + \log \mathcal{N}(x_n | \mu_k, \Sigma_k)) + \sum_{n=1}^N H(r_{nk}) + \lambda^* (\sum_k \pi_k - 1)$$

$$\frac{\partial L}{\partial \pi_k} = \frac{\partial}{\partial \pi_k} \sum_{n=1}^N (\gamma_{nk} \log \pi_k) + \lambda^* \pi_k$$

$$\frac{\partial L}{\partial \pi_k} = \sum_{n=1}^N \gamma_{nk} / \pi_k + \lambda^* = 0$$

$$\pi_k^* = \frac{\sum_n \gamma_{nk}}{\lambda^*}$$

$$\pi_k^* = \frac{\sum_n \gamma_{nk}}{-N}$$

- we know $\sum_k \gamma_{nk} = 1$

$$\frac{\partial L}{\partial \lambda} = \sum_k \pi_k = 1$$

$$\sum_k \pi_k = \sum_k \frac{-\sum_n \gamma_{nk}}{\lambda^*} = 1$$

$$-N/\lambda^* = 1$$

$$\lambda^* = -N$$

[3] EM for GMM (Maximization M step)

- $$\mu_k = \frac{\sum_{n=1}^N \gamma_{nk} x_n}{\sum_{n=1}^N \gamma_{nk}}$$

- M steps for GMM

- $$\Sigma_k = \frac{\sum_{n=1}^N \gamma_{nk} (x_n - \mu_k)^t (x_n - \mu_k)}{\sum_{n=1}^N \gamma_{nk}}$$

- $$\pi_k = \frac{\sum_n \gamma_{nk}}{N}$$

they are the sample means
weighted by responsibility

$$\gamma_{nk} = P[Z_k = k | X_k = x_k, \mu_k(t), \pi_k(t), \Sigma_k(t)]!!$$

[4] EM for GMM (summary)

[1] **E step:** compute the tight lower bound at current parameters

compute $\gamma_{nk} = P[Z_n = k \mid X_n = x_k, \mu_k(t), \pi_k(t), \Sigma_k(t)]$

[2] **M step:** update parameters
using the responsibility

$$\mu_k = \frac{\sum_{n=1}^N \gamma_{nk} x_n}{\sum_{n=1}^N \gamma_{nk}}$$

$$\Sigma_k = \frac{\sum_{n=1}^N \gamma_{nk} (x_n - \mu_k)^t (x_n - \mu_k)}{\sum_{n=1}^N \gamma_{nk}}$$

$$\pi_k = \frac{\sum_n \gamma_{nk}}{N}$$

**** retake the previous example (sprinkler, rain, cloud)**

Let's go back to the problem of
Learning the parameters of Bayesian network.

****** retake the previous example (sprinkler, rain, cloud)

what if data is partially observed?

Suppose we forgot to collect cloud information.

Can we still estimate *p or q* : $P[\text{rain} | \text{cloud} +]$ or $P[\text{rain} | \text{cloud} -]$?

**** retake the previous example (sprinkler, rain, cloud)**

- E step : compute $\gamma_{nk}(t) = P[\text{Cloud}(n) = k \mid \text{Rain}(n) \text{ and } S(n)]$

- M step: update the parameters

- $$P(C)(t+1) = \frac{\sum_{n=1}^N \gamma_{nk}(t)}{N}$$

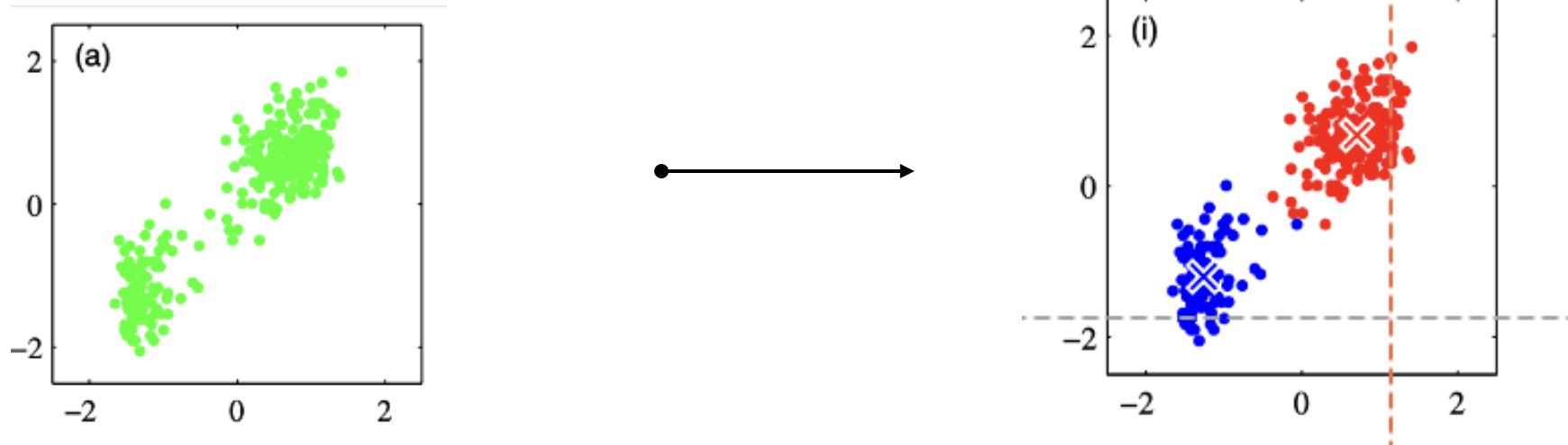
- $$P(S : +, R : + | C : +)(t+1) = \frac{\sum_{n=1}^N \gamma_{nk} \delta(x_n = S : +, R : +)}{\sum_{n=1}^N \gamma_{nk}}$$

- Clustering
partitioned data into a specified number of non-overlapping groups
 - internally similar
 - distinct from each other

- K-means Clustering
parametric and centroid based

[1] K-means Clustering (goal)

[K-means Clustering]:
a clustering method; portioning data into K groups.



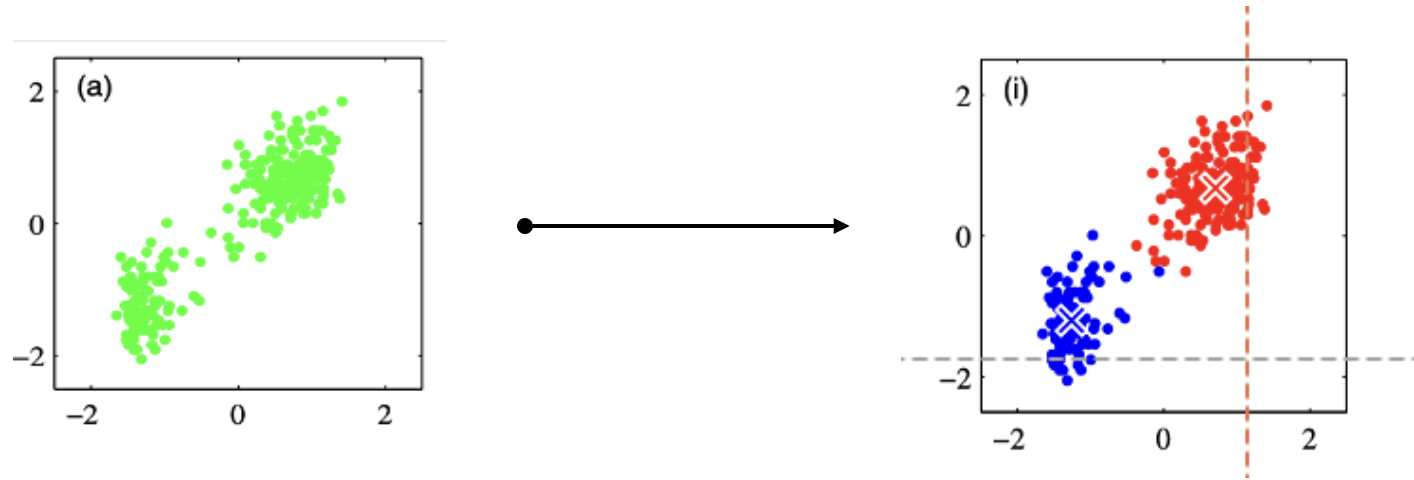
Example) $K=2$

Each data point is mapped to blue / red.

[2] K-means Clustering (goal)

[K-means Clustering]:

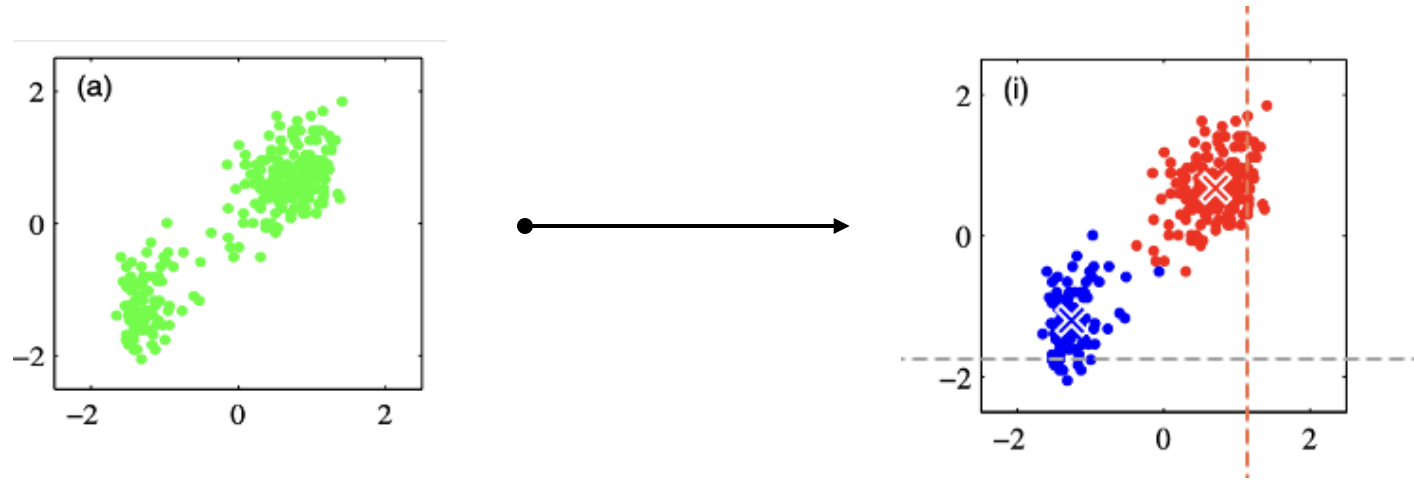
a clustering method; portioning data into K groups.



- K-means algorithm can be understood in the context of GMM and EM, but with approximation & simplification.

[3] K-means Clustering (goal)

[K-means Clustering]:
a clustering method; portioning data into K groups.



- K-means algorithm finally learns $\gamma_{nk} \sim P(Z_b = k \mid x_n)$

[4] K-means Clustering (objective function)

$$J = \sum_{n=1}^N \sum_k \gamma_{nk} \cdot ||x_n - \mu_k||^2 \quad [\text{objective function}]$$

The goal of K-means is to find the optimal values γ_{nk} and μ_k that minimize the objective function.

Q: how can we solve the optimization?

[5] K-means Clustering (objective function)

$$J = \sum_{n=1}^N \sum_k \gamma_{nk} \cdot ||x_n - \mu_k||^2 \quad [\text{objective function}]$$

The goal of K-means is to find the optimal values γ_{nk} and μ_k that minimize the objective function.

Q: how can we solve the optimization?

A: coordinate wise optimization.

[6] K-means Clustering (Coordinate Optimization)

$$J = \sum_{n=1}^N \sum_k \gamma_{nk} \cdot ||x_n - \mu_k||^2 \quad [\text{objective function}]$$

The goal of K-means is to find the optimal values γ_{nk} and μ_k that minimize the objective function.

$$J(\gamma_{nk} , \mu_{k(1)}) \geq J(\gamma_{nk(1)} , \mu_{k(1)}) \geq J(\gamma_{nk(1)} , \mu_{k(2)}) \geq J(\gamma_{nk(2)} , \mu_{k(2)}) \dots$$

[7] K-means Clustering (algorithm, #k given / assumed)

(1) choose some initial values for μ_k

$$J = \sum_{n=1}^N \sum_k \gamma_{nk} \cdot ||x_n - \mu_k||^2$$

(2) E step (μ fixed)

- $\gamma_{nk} = 1$ when $k = \operatorname{argmin}_k ||x_n - \mu_k(t)||^2$
- $\gamma_{nk} = 0$ otherwise

(3) M step (γ fixed)

- $$\mu_k = \frac{\sum_{n=1}^N \gamma_{nk} x_n}{\sum_{n=1}^N \gamma_{nk}}$$

[8] K-means Clustering (comparison of objective functions)

- [GMM Lower Bound]

$$J(\pi, \mu, \Sigma) = \sum_{n=1}^N \sum_k \gamma_{nk} \cdot \left\{ \log \pi_k + \log \frac{1}{\sqrt{(2\pi)^D |\Sigma|}} - 1/2 (x_n - \mu_k)^t \Sigma^{-1} (x_n - \mu_k) \right\} + \sum_{n=1}^N H(\gamma_{nk})$$

- [K-Means Objective Function]

- π_k uniform,
- $\Sigma = I$

$$J = \sum_{n=1}^N \sum_k \gamma_{nk} \cdot \|x_n - \mu_k\|^2$$

Q: do you think the updates by K-means could lead to a monotonic increase in log likelihood ?

[9] K-means Clustering (objective function comparison)

[1] E step

compute $\gamma_{nk} = P[Z_k = k | X_k = x_k, \mu_k(t), \pi_k(t), \Sigma_k(t)]$

[2] M step

- $$\mu_k = \frac{\sum_{n=1}^N \gamma_{nk} x_n}{\sum_{n=1}^N \gamma_{nk}}$$
- $$\Sigma_k = \frac{\sum_{n=1}^N \gamma_{nk} (x_n - \mu_k)^t (x_n - \mu_k)}{\sum_{n=1}^N \gamma_{nk}}$$
- $$\pi_k = \frac{\sum_n \gamma_{nk}}{N}$$

[10] K-means Clustering

[1] E step

compute $\gamma_{nk} = 1$ when $k = \operatorname{argmin}_{\{k\}} P[Z_k = k | X_k = x_k, \mu_k(t), \pi_k(t), \Sigma_k(t)]$
 $\gamma_{nk} = 0$ o.w

$$\text{prob} \propto ||x_n - \mu_k||^{**2}$$

[2] M step

- $$\mu_k = \frac{\sum_{n=1}^N \gamma_{nk} x_n}{\sum_{n=1}^N \gamma_{nk}}$$

- $$\Sigma_k = \frac{\sum_{n=1}^N \gamma_{nk} (x_n - \mu_k)^t (x_n - \mu_k)}{\sum_{n=1}^N \gamma_{nk}}$$

- $$\pi_k = \frac{\sum_n \gamma_{nk}}{N}$$

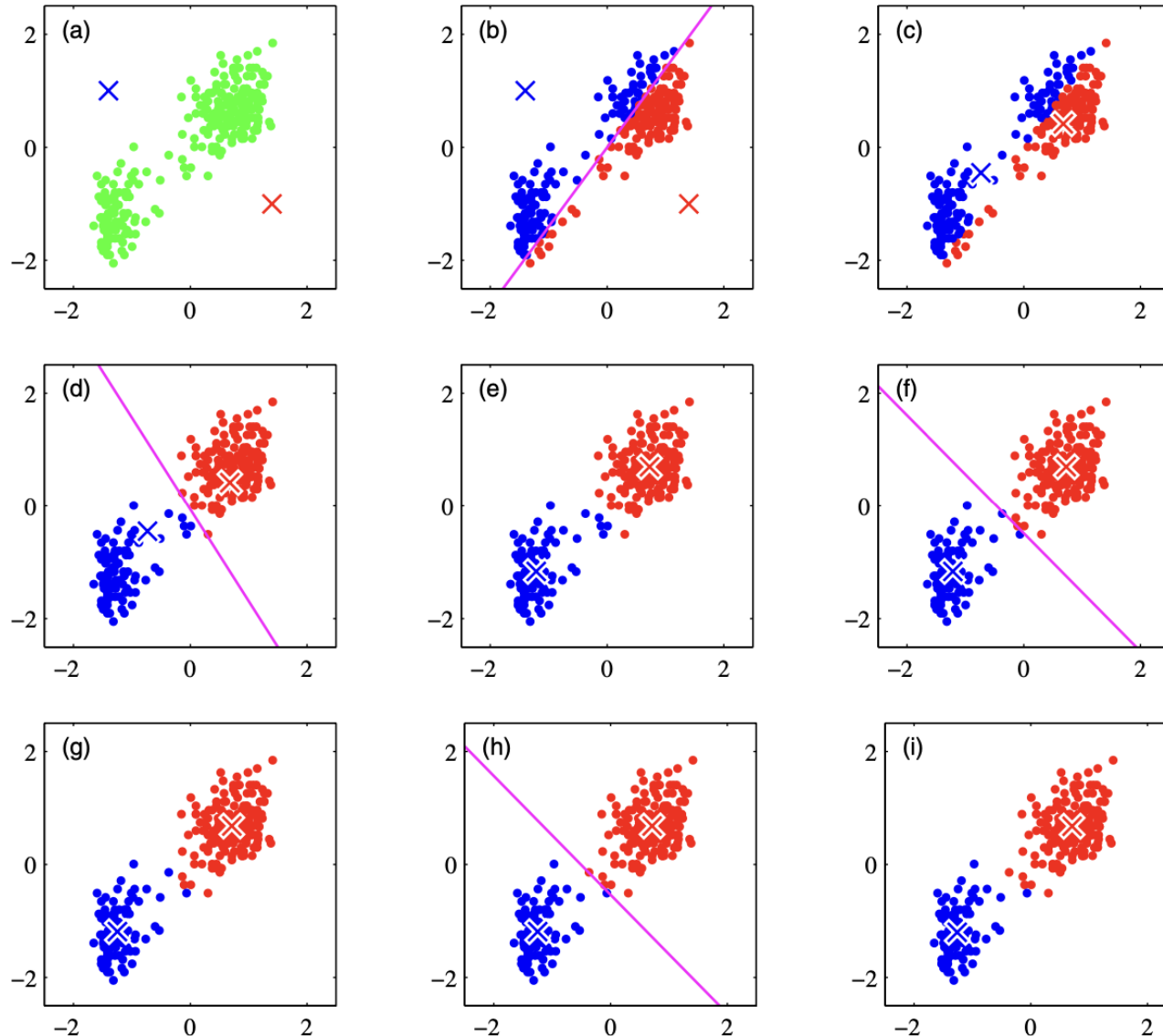
[11] K-means Clustering (comparison between K-means vs. EM for GMM)

GPT comparison😊

Aspect	K-means	EM for GMM
Assignment	Hard	Soft (probabilistic)
Distribution Assumption	Implicit: spherical Gaussians with equal variance	Explicit: full Gaussian (can have different covariances)
Cluster shape	Spherical	Elliptical
Parameters	Only means (centroids)	Means, covariances, and weights

Q: do you think K-mean will show the monotonic increasing behavior?

[12] K-means Clustering (illustration of K-means progression)



****continue until
there is no further changes
in the assignment.**

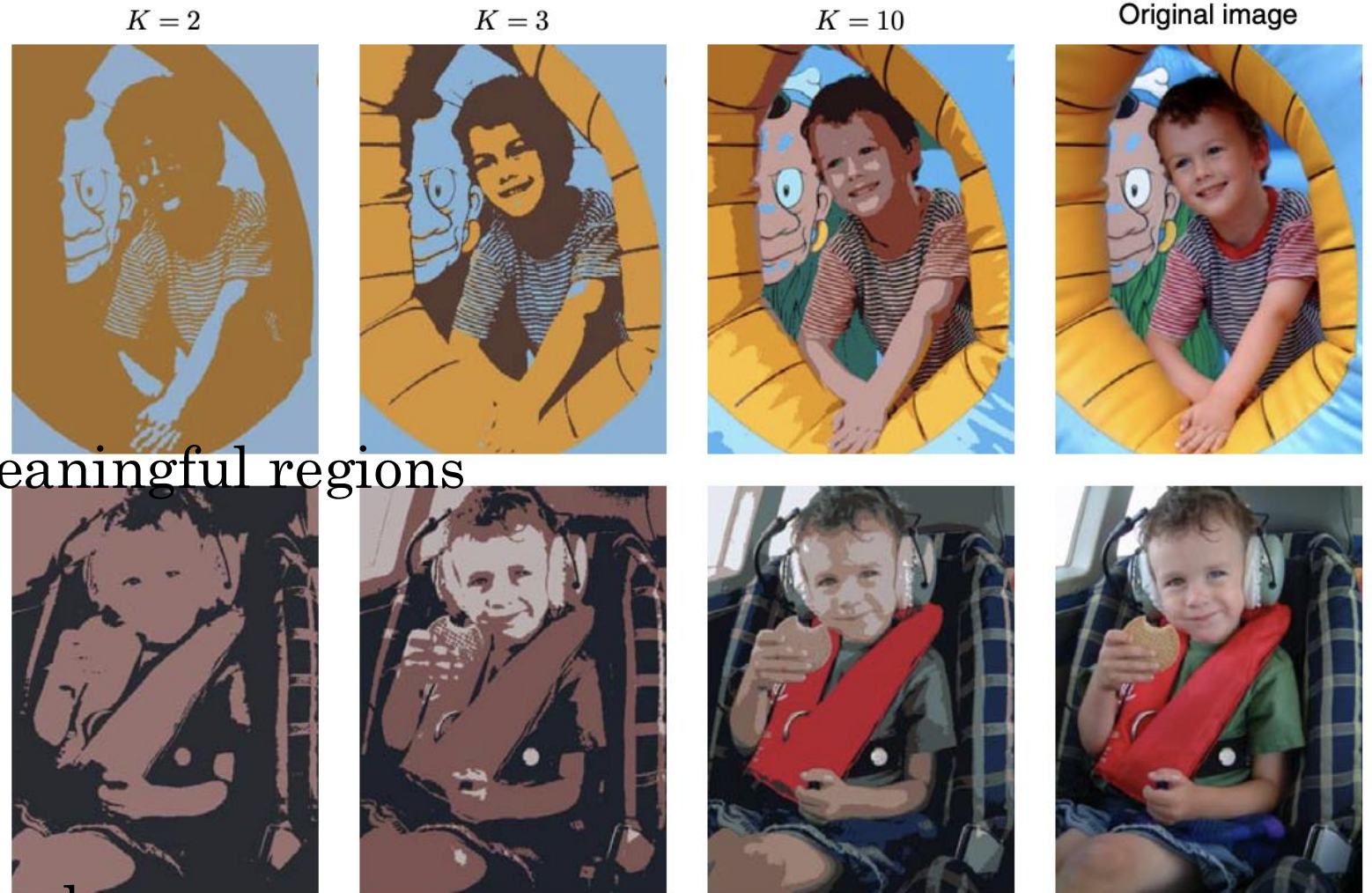
from textbook Bishop Fig. 9.1

[13] K-means Clustering (K-means examples)

- image segmentation
- image compression

**labeling
partitions the image into meaningful regions
not sophisticated one, why?

**compression
images with fewer distinct colors



[14] K-means Clustering (limitation)

- $\gamma_{nk} = 1$ when $k = \operatorname{argmin}_k ||x_n - \mu_k(t)||^2$
- $\gamma_{nk} = 0$ otherwise

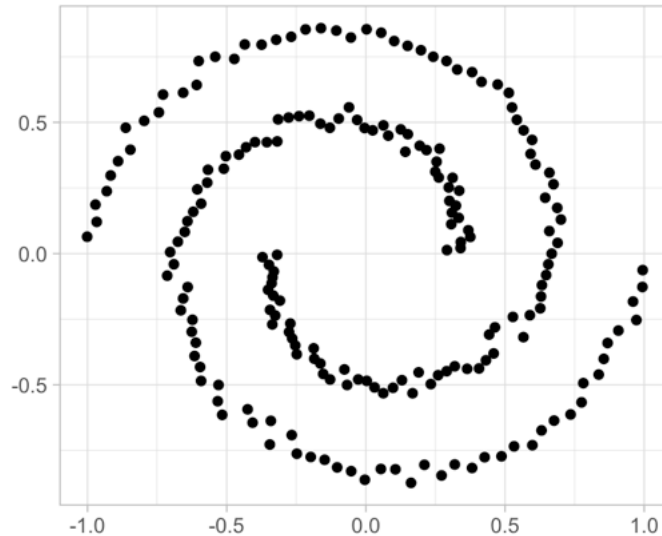
Q: What if the data has a connected manifold and apply K means ($K = 2$)?

[15] K-means Clustering (limitation)

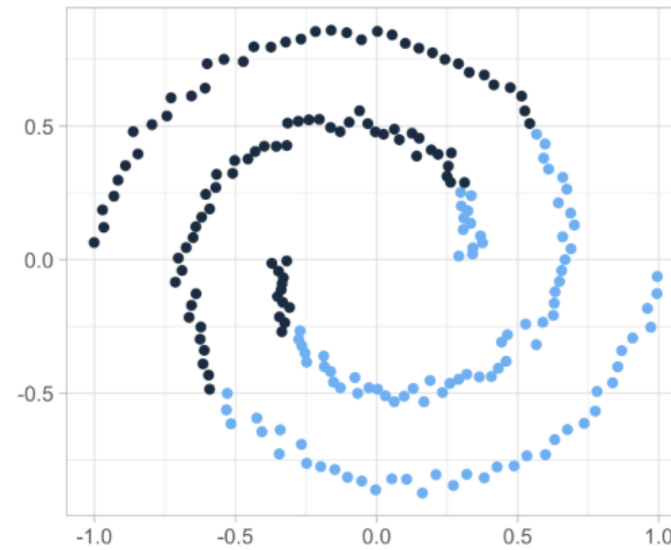
- $\gamma_{nk} = 1$ when $k = \operatorname{argmin}_k ||x_n - \mu_k(t)||^2$
- $\gamma_{nk} = 0$ otherwise

Q: What if the data has a connected manifold and apply K means (K = 2)?

(a) Original data



(b) K means Cluster



<https://bradleyboehmke.github.io/HOML/kmeans.html>

[16] K-means Clustering (limitation)

- (1) vulnerable when data lies on a connected or non-linearly separable manifold.
(for Gaussian assumption)
- (2) predefined K
- (3) sensitive to outlier.

- DBSCAN, non-parametric & density based
(Density Based Spatial Clustering of Application with Noise)

[paper review]

<https://www.dbs.ifi.lmu.de/Publikationen/Papers/KDD-96.final.frame.pdf>

[1] DBSCAN (eps-neighborhood)

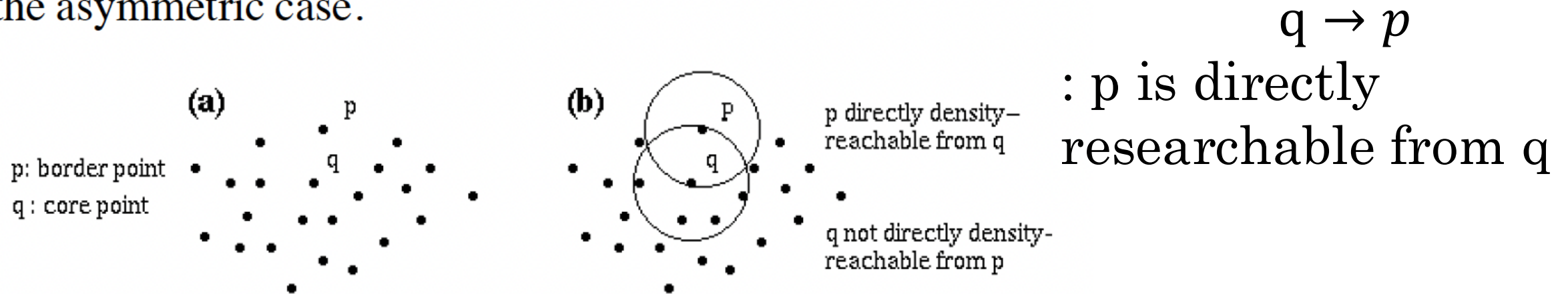
Definition 1: (Eps-neighborhood of a point) The *Eps-neighborhood* of a point p , denoted by $N_{\text{Eps}}(p)$, is defined by $N_{\text{Eps}}(p) = \{q \in D \mid \text{dist}(p,q) \leq \text{Eps}\}$.

[2] DBSCAN (directly density reachable)

Definition 2: (directly density-reachable) A point p is *directly density-reachable* from a point q wrt. Eps , $MinPts$ if

- 1) $p \in N_{Eps}(q)$ and
- 2) $|N_{Eps}(q)| \geq MinPts$ (core point condition).

Obviously, directly density-reachable is symmetric for pairs of core points. In general, however, it is not symmetric if one core point and one border point are involved. Figure 2 shows the asymmetric case.



[3] DBSCAN (directly density reachable)

Definition 3: (density-reachable) A point p is *density-reachable* from a point q wrt. Eps and MinPts if there is a chain of points $p_1, \dots, p_n$, $p_1 = q$, $p_n = p$ such that p_{i+1} is directly density-reachable from p_i .

$q \rightarrow p_1 \rightarrow p_2 \dots p_{n-1} \rightarrow p$
(core points)
: p is researchable from q

[4] DBSCAN (density connected)

Definition 4: (density-connected) A point p is *density-connected* to a point q wrt. Eps and $MinPts$ if there is a point o such that both, p and q are density-reachable from o wrt. Eps and $MinPts$.

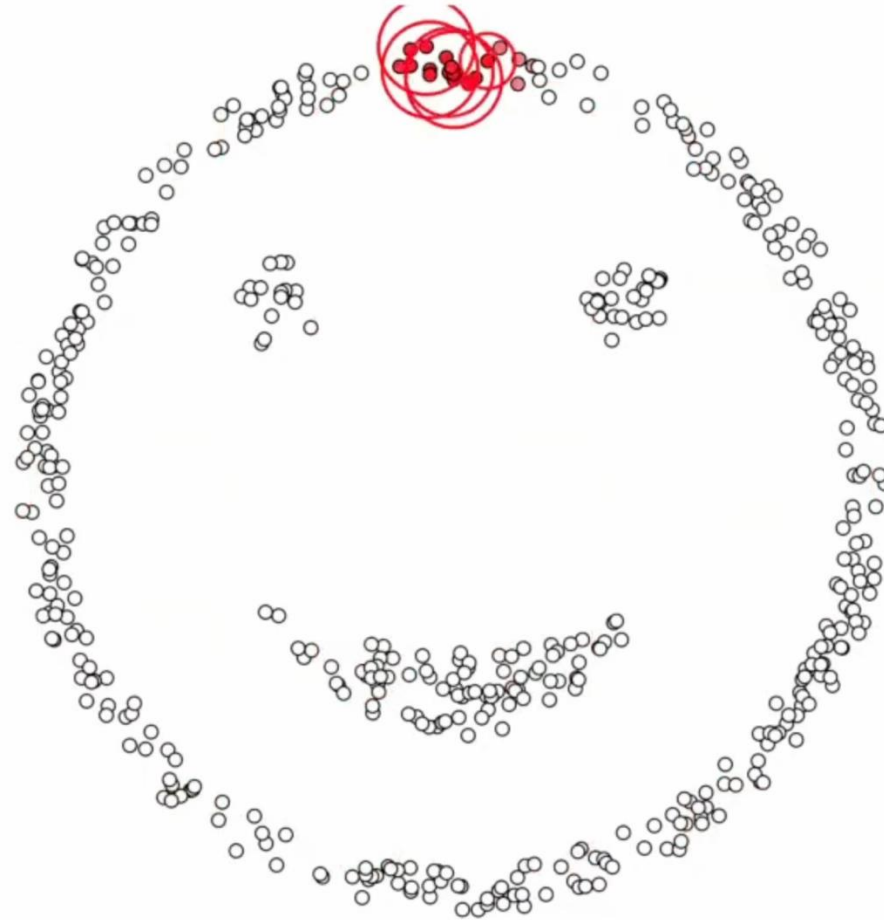
$p \rightarrow o \ \& \ q \rightarrow o$

: density connected (p and q are the same cluster)

[5] DBSCAN (algorithm)

1. Start with a core point “ p ” and form a new cluster.
2. All eps neighbors (p) are added to the cluster.
3. All eps neighbors (p) are marked as visited.
4. Expand the cluster by adding the neighbors of each of eps neighbors (p).
5. Until no more point to expand.
6. Explore another core points not visited before.
7. Any not clustered points become noise.

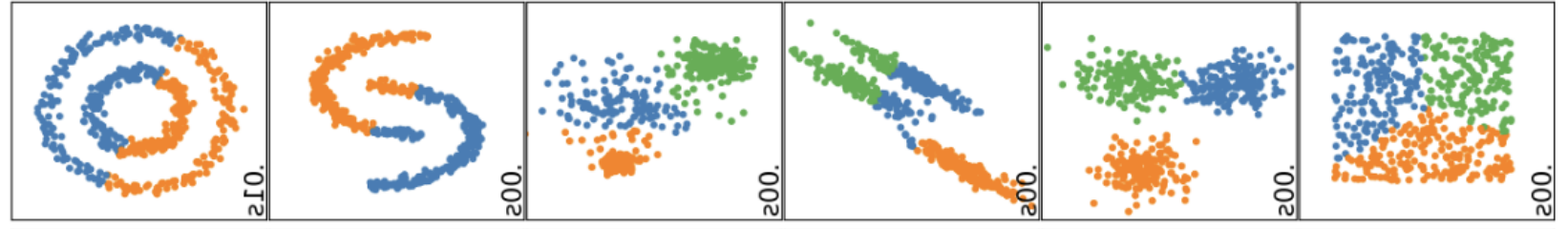
[6] DBSCAN (example)



https://www.youtube.com/watch?v=_A9Tq6mGtLI&t=3s

[7] DBSCAN (results)

[K means]



[DBSCAN]

